germ cells by a long series of mitoses and which are in different stages of differentiation. **Sperm** (spermatozoa) are formed only after sexual maturity is reached. The last cell type in the developmental sequence is the **primary spermatocyte,** a diploid germ cell that undergoes meiosis I to form two haploid **secondary spermatocytes**. Secondary spermatocytes rapidly enter meiosis II, each forming two **spermatids,** which differentiate without further division into sperm. In humans, the entire process takes approximately 64 days. The enormous number of sperm produced, typically approximately 200 million per ejaculate and an estimated $10^{12}$ in a lifetime, requires several hundred successive mitoses.

As discussed earlier, normal meiosis requires pairing of homologous chromosomes followed by recombination. The autosomes and the X chromosomes in females present no unusual difficulties in this regard; but what of the X and Y chromosomes during spermatogenesis? Although the X and Y chromosomes are different and are not homologues in a strict sense, they do have relatively short identical segments at the ends of their respective short arms (Xp and Yp) and long arms (Xq and Yq) (see Chapter 6). Pairing and crossing over occurs in both regions during meiosis I. These homologous segments are called **pseudoautosomal** to reflect their autosome-like pairing and recombination behavior, despite being on different sex chromosomes.

## Oogenesis

Whereas spermatogenesis is initiated only at the time of puberty, oogenesis begins during a female's development as a fetus (Fig. 2-17). The **ova** develop from **oogonia,** cells in the ovarian cortex that have descended from the primordial germ cells by a series of approximately 20 mitoses. Each oogonium is the central cell in a developing follicle. By approximately the third month of fetal development, the oogonia of the embryo have begun to develop into **primary oocytes,** most of which have already entered prophase of meiosis I. The process of oogenesis is not synchronized, and both early and late stages coexist in the fetal ovary. Although there are several million oocytes at the time of birth, most of these degenerate; the others remain arrested in prophase I (see Fig. 2-14) for decades. Only approximately 400 eventually mature and are ovulated as part of a woman's menstrual cycle.

After a woman reaches sexual maturity, individual follicles begin to grow and mature, and a few (on average one per month) are ovulated. Just before ovulation, the oocyte rapidly completes meiosis I, dividing in such a way that one cell becomes the secondary oocyte (an egg or **ovum**), containing most of the cytoplasm with its organelles; the other cell becomes the first polar body (see Fig. 2-17). Meiosis II begins promptly and proceeds to the metaphase stage during ovulation, where it halts again, only to be completed if fertilization occurs.

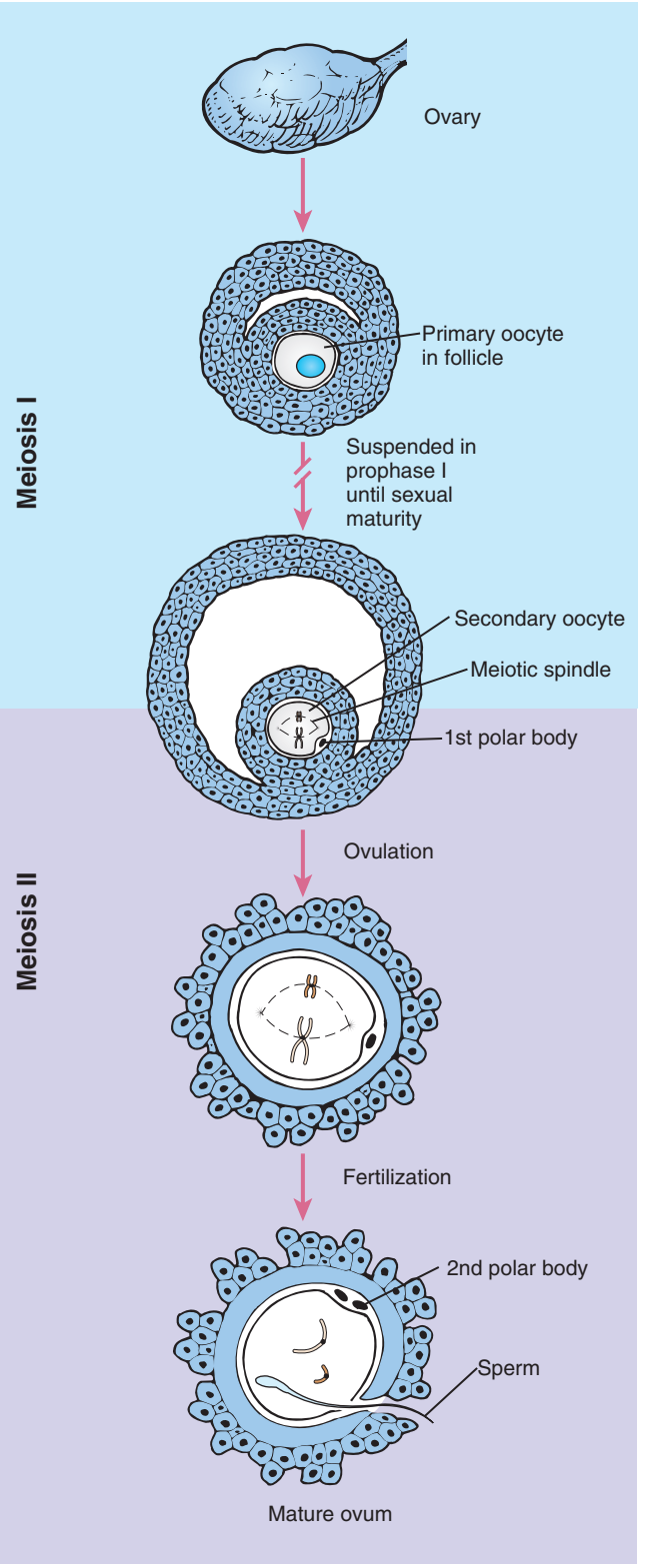



**Figure 2-17 Human oogenesis and fertilization in relation to the two meiotic divisions.** The primary oocytes are formed prenatally and remain suspended in prophase of meiosis I for years until the onset of puberty. An oocyte completes meiosis I as its follicle matures, resulting in a secondary oocyte and the first polar body. After ovulation, each oocyte continues to metaphase of meiosis II. Meiosis II is completed only if fertilization occurs, resulting in a fertilized mature ovum and the second polar body.